
Belief-Based Personalization for Menstrual Cycle Phase Tracking Using Wearable Data

Alina Gavrilov
Stanford University
alinagav@stanford.edu

Abstract

Predicting menstruation onset from wearable data remains challenging because the underlying hormonal phase is unobserved and symptoms are sparsely reported. I develop a proof-of-concept framework that models the menstrual cycle using a hidden semi-Markov model (HSMM) (3) and formulates personalized onset prediction as a partially observable Markov decision process (POMDP) (1) with actions for waiting, querying symptoms, issuing alerts, and recommending self-care. Beliefs are updated using daily biometric and survey-derived risk signals, and policies are evaluated using both a hand-designed threshold rule and a point-based value iteration (PBVI) approximation. In this initial implementation, neither method produced timely alerts, highlighting limitations in the current emission model, reward shaping, and action effects. Nonetheless, the results demonstrate that a generative-model-based POMDP provides a principled structure for sequential menstrual inference, and they point toward clear directions for improvement using richer survey data, multi-output physiological models, and more expressive planners.

1 Project Description: Menstrual Cycle, Wearable Tech, and POMDPs

1.1 Background on Menstrual Cycle Predictions

The menstrual cycle consists of four phases—menstrual, follicular, ovulation, and luteal—but their durations vary widely across individuals. Research shows that cycle variability is influenced by stress, hormonal fluctuations, athletic training, and reproductive health conditions (6). Despite this, most commercial apps and wearables predict cycle timing using only historical averages of past bleeding dates. This approach fails for the many menstruating people who experience irregular cycles, including those with PMS, PCOS, endometriosis, fibroids, and other conditions.

Biometric signals captured by wearables (7), especially when combined with rich symptom survey data, provide a far more informative view of the underlying physiology. Accurate phase prediction can improve daily planning, support fertility tracking, and help identify patterns associated with reproductive disorders. My research group at Stanford maintains an ongoing study in which participants wear Oura Rings and complete weekly symptom surveys, creating a unique multimodal dataset for personalized inference. By combining an HSMM generative model of realistic phase durations with a POMDP framework for sequential decision-making under uncertainty, the method can maintain beliefs over latent menstrual states and choose information-gathering or alerting actions that adapt to each user’s physiology.

1.2 POMDP Structure

In this project, I introduce a POMDP formulation for menstrual phase tracking, treating the true phase as a latent state that must be inferred from noisy wearable signals and sparse symptom reports

(Table 2) (1). I evaluate two policy search approaches for determining when the system should wait, query symptoms, alert a user to a predicted menstruation onset, or recommend a care-oriented action. The CARE action is an abstraction for behaviors such as resting, reducing stress, or adjusting activity levels—actions known to modulate the likelihood of experiencing negative symptoms around menstruation (Table 2). Including CARE allows the agent to influence the latent “risk” component of the state, which represents susceptibility to symptoms like cramps, fatigue, mood changes, and other discomforts commonly associated with the bleeding phase. A generative Hidden Semi-Markov Model (HSMM) specifies both how latent states (phase and risk) evolve over time and how observations (biometrics and symptoms) are produced from those states. Using this model, I evaluate a threshold policy and PBVI to balance my two objectives: accurate onset prediction and lowering symptom risk.

State	Meaning
M-H	Menstruation, high risk
M-L	Menstruation, low risk
F-H	Follicular, high risk
F-L	Follicular, low risk
O-H	Ovulation, high risk
O-L	Ovulation, low risk
L-H	Luteal, high risk
L-L	Luteal, low risk

Table 1: *

Latent states (phase $\times$ risk)

Action	Effect
WAIT	Collect wearables only
QUERY	Request symptom report
ALERT	Predict/announce onset
CARE	Recommend self-care to reduce risk

Table 2: *

Actions

2 Data Processing

Daily physiological HRV and temperature features (4) were extracted from raw Oura Ring data using a standardized preprocessing pipeline from our lab. For HRV, I identified each participant’s longest sleep period per night and used the corresponding nighttime average HRV value. Temperature deviation was obtained via a stable sliding window run on Oura’s nightly temperature signal. Both HRV and temperature were log-transformed and then smoothed using single-output Gaussian process regression to reduce noise and interpolate occasional missing days.

Each survey contains two sets of binary indicators: physical symptoms (e.g., cramping, headache, trouble sleeping) and mental symptoms (e.g., anxiety, difficulty concentrating, mood swings). “None” fields were removed, missing values were set to zero, and all remaining symptom variables were cast to binary.

For each survey, three summary quantities were computed: **physical_count**: total number of physical symptoms reported, **mental_count**: total number of mental symptoms reported, **functional_flag**: whether participant reports taking day off due to symptoms.

These were combined into a single daily **risk observation** defined as

$$\text{risk_obs} = \begin{cases} 1 & \text{if functional_flag} = 1, \\ 1 & \text{if physical_count} \geq 2, \\ 1 & \text{if mental_count} \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Each weekly survey interval was then expanded into daily rows covering every date from its reported start to end date. The resulting daily dataset assigns each participant a binary risk flag for each day, along with one-day indicators for recorded menstruation or ovulation events. The resulting dataset contains, for each participant and each day, a smoothed HRV value, a smoothed temperature deviation, a risk flag, menstruation flag, and ovulation flag. These daily flags serve as the training targets for estimating the phase-conditioned risk transition model. These daily features serve as the observable inputs to the generative model and belief update process.

3 Generative Model

The menstrual cycle is modeled using a hidden semi-Markov model (HSMM) that captures both the canonical ordering of hormonal phases and their typical durations (3).

3.1 Transition Model

Each phase follows the physiological progression $M \rightarrow F \rightarrow O \rightarrow L \rightarrow M$, and the HSMM provides learned dwell-time distributions rather than memoryless transitions. These produce semi-Markov transition probabilities $P(P_{t+1} | P_t)$ that reflect realistic variability in how long individuals remain in each phase.

On top of this phase progression, the generative model includes a second latent variable, $\text{Risk}_t \in \{0, 1\}$, indicating low or high likelihood of experiencing symptoms. Risk evolves stochastically within each phase according to empirical transitions learned from day-to-day symptom data:

$$P(\text{Risk}_{t+1} | \text{Risk}_t, P_t).$$

The CARE action modifies these risk transitions by increasing the probability of switching from high to low risk, while phase dynamics remain uncontrollable. All other actions leave the baseline risk dynamics unchanged.

Combining phase and risk yields an eight-state latent space

$$s_t = (P_t, \text{Risk}_t).$$

The full transition model factorizes as

$$P(s_{t+1} | s_t, a_t) = P(P_{t+1} | P_t) P(\text{Risk}_{t+1} | \text{Risk}_t, P_t, a_t),$$

where a_t = CARE selects the CARE-modified risk transition and all other actions use the baseline risk model. Observations depend only on the current state, implying the conditional independence structure

$$P(o_t | s_t, s_{t-1}, a_{t-1}) = P(o_t | s_t).$$

3.2 Observation Model

Each day the agent receives continuous biometric observations

$$X_t = (\text{temp}_t, \text{hrv}_t),$$

modeled as a Gaussian emission conditioned only on phase:

$$X_t | (P_t = p, \text{Risk}_t = r) \sim \mathcal{N}(\mu_p, \Sigma_p).$$

These parameters are taken from the HSMM fit to the wearable data.

When the agent selects QUERY, it additionally observes a binary symptom variable

$$Y_t | \text{Risk}_t = r \sim \text{Bernoulli}(p_r),$$

where p_r is the empirically estimated probability of reporting symptoms under risk level r . Symptoms depend on risk but not on phase. Missing symptom observations (i.e., when not queried) are treated as uninformative.

4 POMDP Policy Selection

4.1 Belief Updates

Let $b_t(s)$ denote the belief over states. The prediction step applies the transition model:

$$\hat{b}_t(s') = \sum_s b_{t-1}(s) P(s' | s, a_{t-1}).$$

Observations are incorporated through Bayes' rule (1): $b_t(s') \propto \hat{b}_t(s') P(o_t | s')$. The implementation uses log-space likelihoods for numerical stability and normalizes via a softmax.

4.2 Reward Model

The reward function (1) reflects three objectives: minimizing user burden, identifying high-risk periods, and producing timely onset alerts. The per-action rewards are

$$R(\text{WAIT}) = 0, \quad R(\text{QUERY}) = -1, \quad R(\text{CARE}) = -1.$$

The reward for ALERT depends on the timing error $\Delta = \text{day}_{\text{alert}} - \text{day}_{\text{true onset}}$:

$$R(\text{ALERT}) = \begin{cases} 10 & |\Delta| \leq 1, \\ 3 & 2 \leq |\Delta| \leq 3, \\ -5 & \Delta < -3, \\ -10 & \Delta > 1. \end{cases}$$

Thus the policy is rewarded for predicting onset within a narrow window and penalized for late or substantially early alerts.

4.3 Threshold Policy

As a simple interpretable baseline, I implement a threshold policy operating directly on the belief distribution. Let

$$P_{\text{risk}} = P(R_t = 1 \mid b_t), \quad P_{\text{late}} = P(P_t = L \mid b_t)$$

denote the marginal high-risk and late-luteal probabilities. The policy selects

$$\pi(b_t) = \begin{cases} \text{CARE}, & P_{\text{risk}} > \tau_{\text{risk}}, \\ \text{ALERT}, & P_{\text{late}} > \tau_{\text{onset}}, \\ \text{QUERY}, & H(b_t) > \tau_{\text{ent}}, \\ \text{WAIT}, & \text{otherwise}, \end{cases}$$

where $H(b) = -\sum_s b(s) \log b(s)$ is the belief entropy. Thus, CARE is issued when high risk is likely, ALERT when onset appears imminent, QUERY when uncertainty is high, and WAIT otherwise.

4.4 Threshold Policy with Hyperparameter Search

Because the threshold policy's behavior depends strongly on its three parameters, I performed a grid search over

$$\tau_{\text{risk}} \in \{0.30, 0.35, 0.40, 0.45, 0.50\}, \quad \tau_{\text{onset}} \in \{0.15, 0.20, 0.25, 0.30\}, \quad \tau_{\text{ent}} \in \{1.0, 1.1, 1.2, 1.3, 1.4\}.$$

Each configuration was evaluated on the complete test set. This yielded

$$(\tau_{\text{risk}}^*, \tau_{\text{onset}}^*, \tau_{\text{ent}}^*) = (0.30, 0.15, 1.0),$$

the setting used for the threshold-policy results reported in the Results section. These threshold ranges were chosen based on exploratory belief trajectories in the training set, where lower values triggered excessive actions and higher values suppressed alerts entirely.

4.5 Point-Based Value Iteration (PBVI)

As a second baseline, I implemented a lightweight point-based value iteration (PBVI) method to approximate an optimal policy over the belief space (2). A fixed set of belief points was used, and for each point the algorithm performs a sample-based backup: next states and observations are generated from the HSMM-based transition and emission model, the belief is updated via the Bayesian filter, and the resulting value estimate is used to update the corresponding α -vector. Because the reward for ALERT depends on timing error, which is not defined inside the backup operator, the PBVI implementation optimizes a simplified reward structure in which ALERT yields zero immediate reward.

After 8 iterations, the resulting PBVI policy was evaluated on the test set. In practice, it failed to produce any meaningful onset alerts and did not outperform the threshold policy on any metric. The simplified reward structure and limited number of belief points prevented PBVI from capturing the sharp timing behavior required for accurate menstruation-onset prediction.

4.6 Evaluation Metrics

For each cycle, the timing error

$$\Delta = \text{day}_{\text{alert}} - \text{day}_{\text{true onset}}$$

is computed. From these, the following metrics are reported:

- **Timely alert rate:** fraction of alerts with $|\Delta| \leq 1$.
- **False-alarm rate:** fraction of alerts issued more than 3 days early.
- **Miss/late rate:** cycles with no alert or with a late alert ($\Delta > 1$).
- **Average queries:** mean number of QUERY actions per cycle.
- **Average reward:** mean cumulative reward across cycles.

These metrics capture accuracy, conservativeness, and user burden.

5 Threshold Policy Results

Tables 3 and 4 summarize the performance of the threshold policy on the complete and partial test sets. Across both datasets, the policy almost never triggers timely alerts, produces very large timing errors when it does alert, and accumulates substantial negative reward due to frequent querying. These results confirm that the hand-designed thresholding rules are insufficient for robust onset prediction. In Table 5 you see the threshold search. False-alarm rates were low for many settings because the model seldom issued any alerts. Improvements in reward were driven primarily by issuing fewer queries, not better onset prediction. The best parameter set under the false-alarm constraint was (0.3, 0.15, 1.0), but it still failed to generate correct menstrual onset alerts. Zero values here reflect the policy’s failure to accumulate sufficient belief to trigger any action, not missing data.

Metric	Value
Timely alert rate ($ \Delta \leq 1$)	0.00
False alarm rate ($\Delta < -3$)	0.00
Miss/late rate	1.00
Average number of queries	5.71
Average reward	−141.86
Timing error mean Δ	37.50
Timing error std	33.64
Min / Median / Max Δ	9 / 21 / 81

Table 3: Threshold policy performance on the complete test set.

Metric	Value
Timely alert rate ($ \Delta \leq 1$)	0.00
False alarm rate ($\Delta < -3$)	0.05
Miss/late rate	0.95
Average number of queries	19.05
Average reward	−763.52
Timing error mean Δ	64.70
Timing error std	75.65
Min / Median / Max Δ	3 / 30 / 217

Table 4: Threshold policy performance on the partial test set.

Selected configuration results from Table 5: $\tau_{\text{risk}} = 0.30$, $\tau_{\text{onset}} = 0.15$, $\tau_{\text{ent}} = 1.0$; timely alert rate 0.00, false-alarm rate 0.00, miss/late rate 1.14, avg. queries 3.71, avg. reward −42.7.

<i>Top candidates (ranked by timely alert rate)</i>					
τ_{risk}	τ_{onset}	τ_{ent}	Timely	False Alarm	Avg. Queries
0.30	0.15	1.0	0.00	0.00	3.71
0.30	0.15	1.1	0.00	0.00	0.00
0.30	0.15	1.2	0.00	0.00	0.00
0.30	0.15	1.3	0.00	0.00	0.00
0.30	0.15	1.4	0.00	0.00	0.00

Table 5: **Threshold-policy hyperparameter sweep.** Across all 100 combinations of $(\tau_{\text{risk}}, \tau_{\text{onset}}, \tau_{\text{ent}})$, the timely alert rate remained zero.

6 Discussion and Conclusion

This project framed menstrual-onset prediction as a POMDP using an HSMM generative model and multimodal wearable–survey data. Although the framework is principled, neither the threshold policy nor the PBVI approximation produced timely alerts. In practice, beliefs rarely accumulated enough probability in the late-luteal region to trigger ALERT, and PBVI did not improve performance.

Several modeling limitations help explain these outcomes. The HSMM emissions and single-output GP smoothing generate signals that are too smooth to distinctively indicate phase transitions. The sharp timing penalty in the onset reward also makes the planning problem difficult for approximate value methods. A smoother, distance-based reward that scales with the temporal proximity of the alert to true onset would make the signal less brittle and promote gradual improvements. Finally, action effects are weak: CARE modifies only risk, and QUERY does not reduce future uncertainty in the model.

Despite these issues, the POMDP formulation remains promising. I now have access to substantially more survey data, which will allow richer learning of symptom dynamics and risk transitions. Future work will focus on improving the HSMM with multi-output biometrics, refining reward shaping for onset timing, and experimenting with stronger solvers such as SARSOP or POMCP. This project served as a proof of concept for structuring menstrual prediction as a sequential decision problem and highlighted the key modeling components that require refinement.

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